Training supervision and deployed control loop

TRAINING ONLY

Privileged simulation
terrain height
true contact force

DLS teacher
$\Delta x^* = \text{clip}(g\ J^{\#}\ W\ e)$
cylinders 1–4; C4 held

DAgger labels
teacher execution
1 → 0

supervision

DEPLOYMENT

Observation
o_k
measurable

PPO actor
$\pi\theta(o_k)$
no nominal input



desired cylinder motion

Actuator path
R | IF | AF
bounded target



PhysX arm plant

measured state closes the deployed loop